

IndoLepAtlas: A Fine-Grained Dataset of Indian Lepidoptera with Geotemporal Metadata and Baseline Classification Study

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Abstract. India is home to an extensive range of insect species but is severely underrepresented in global biodiversity datasets. This creates a significant lack of quality training assets for creating region-specific classification architectures. For species of order Lepidoptera (Butterflies & Moths), with visually striking features that change with geography and climate, the problem is even more pronounced. To mitigate this, we created the IndoLepAtlas, a curated 60,641-image, 967-species **India-specific butterfly** dataset with geotemporal metadata, early life-stage documentation (9,804 images), and 703 images of larval host plant species. The dataset is curated from images submitted to a community-driven source, and is verified over multiple stages, ensuring quality and diversity in butterfly species. Our dataset was also cross-referenced against iNaturalist, a globally renowned biodiversity dataset, revealing a significant butterfly species coverage gap, which IndoLepAtlas intends to bridge. Fine-grained classification of butterfly species, often with very subtle inter-species differences or even great intra-species differences, poses a compounding problem. Our research leveraged a unique deep-learning model with Coordinate Attention (CA), Multi-Level Feature Interaction (MLFI) and geotemporal feature integration. We also demonstrate the effectiveness of our model in accurately classifying butterfly species, showcasing improvements over other commonly used architectures.

Keywords: Butterflies, Fine-grained Visual Classification, Lepidoptera, Indian Biodiversity, Geotemporal Fusion, Long-tail Distribution, Coordinate Attention

1 Introduction

India’s 1,500+ Lepidoptera species represent one of the world’s richest butterfly biodiversity hotspots. Accurate butterfly species classification from field photographs is crucial for tracking, monitoring, research and conservation planning. This task, though, fundamentally differs from standard benchmarks because of a variety of reasons like, discriminative features being spatially precise (eyespot location separates genera), hierarchical distribution across scales (texture bands

vs overall wing morphology), large variations within a class, small variations between different classes and uneven sampling (dominant species exceed 500 images while 400 rare species have <50 images).

Existing global biodiversity datasets—iNaturalist [12], GBIF—dilute India-specific subspecies representation. Many Indian endemic and regionally restricted species have fewer than 100 research-grade observations in global repositories, creating a critical data gap for conservation-focused AI systems. Furthermore, no existing butterfly classification dataset integrates geotemporal metadata (biogeographic zone, observation season) or documents larval host plant associations—ecological dimensions that could substantially improve classification accuracy and enable applications beyond species identification, such as early-stage detection and habitat suitability mapping.

Standard CNNs fail on this task due to *spatial information collapse*, *progressive texture loss*, *long-tail imbalance* among other reasons.

We make the following contributions:

- (1) **IndoLepAtlas dataset:** 60,641 butterfly images, 967 species, GPS coordinates, observation dates, 9-zone biogeographic mapping, early life-stage documentation (9,804 images), and 703 host plant images—all centered around Indian Lepidoptera species.
- (2) **Systematic baseline establishment:** 11-configuration ablation + 6-model comparison providing reproducible baselines.
- (3) **Geotemporal fusion:** Integration of biogeographic zone and seasonal embeddings into butterfly classification pipeline, achieving 89.38% Top-1 with $2.1 \times$ fewer parameters than ViT-B/16, and verified via geo-shuffled ablation.

2 Related Work

2.1 Fine-Grained Visual Classification

Fine-grained visual classification (FGVC) aims to distinguish subordinate categories within a superclass—bird species [11], dog breeds, car models. Key benchmarks include CUB-200-2011 [11] (200 bird species, 11,788 images) and iNaturalist [12] (5,000+ species, 675,000 images). Architecturally, the field has progressed from bilinear pooling through channel attention (SE-Net [4]) to multi-level feature fusion (FGBNet [1]). Vision Transformers [9] achieve strong results on large-scale FGVC but require substantial data. Coordinate Attention [2] offers a lightweight alternative that preserves spatial structure—a property we exploit for wing-pattern localisation.

2.2 Insect Classification Datasets and Systems

The systematic literature review by Amarathunga et al. [13] surveyed insect image classification comprehensively: only 10 of 28 insect orders covered, no Africa or South Asia representation, no morphologically similar microscopic

species datasets, and temporal distribution flagged as an explicit open problem. The thrips classification work [6] is the most relevant morphological similarity study—domain-knowledge-driven body-segment augmentation with ViT achieved 94.7% on microscopic insects, analogous to our challenges. Alfatemi et al. [14] established methodology for manual data curation in visually similar bird species, a protocol we adapt for Lepidoptera.

2.3 Geotemporal Context in Species Recognition

The SLR [13] explicitly identifies temporal and geographic distribution as open problems. Our geotemporal fusion module directly addresses this gap. In addition to geotemporal context, the dataset also provides images of larval stages and larval host plants for further classification enhancements or behaviour modelling.

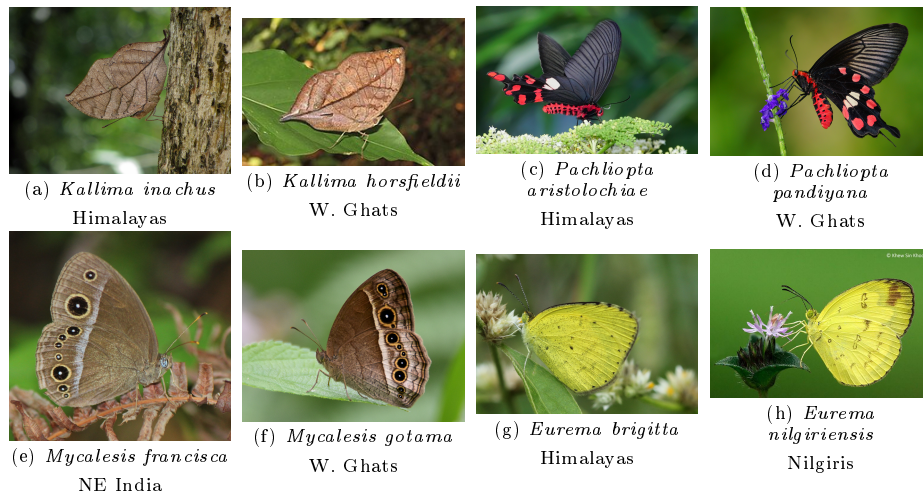


Fig. 1. Four visually near-identical species pairs separated by biogeographic zone. (a–b) *Kallima* dead-leaf butterflies: Himalayas vs. Western Ghats. (c–d) Swallowtail roses: widespread Common Rose vs. Western Ghats–endemic Malabar Rose. (e–f) *Mycalesis* bush browns: Northeast India vs. Western Ghats. (g–h) *Eurema* grass yellows: widespread *E. brigitta* vs. Nilgiri–endemic *E. nilgiriensis*. Geographic metadata disambiguates these pairs where visual features alone cannot.

3 IndoLepAtlas Dataset

The dataset is available at [IndoLepAtlas \(Hugging Face\)](#).

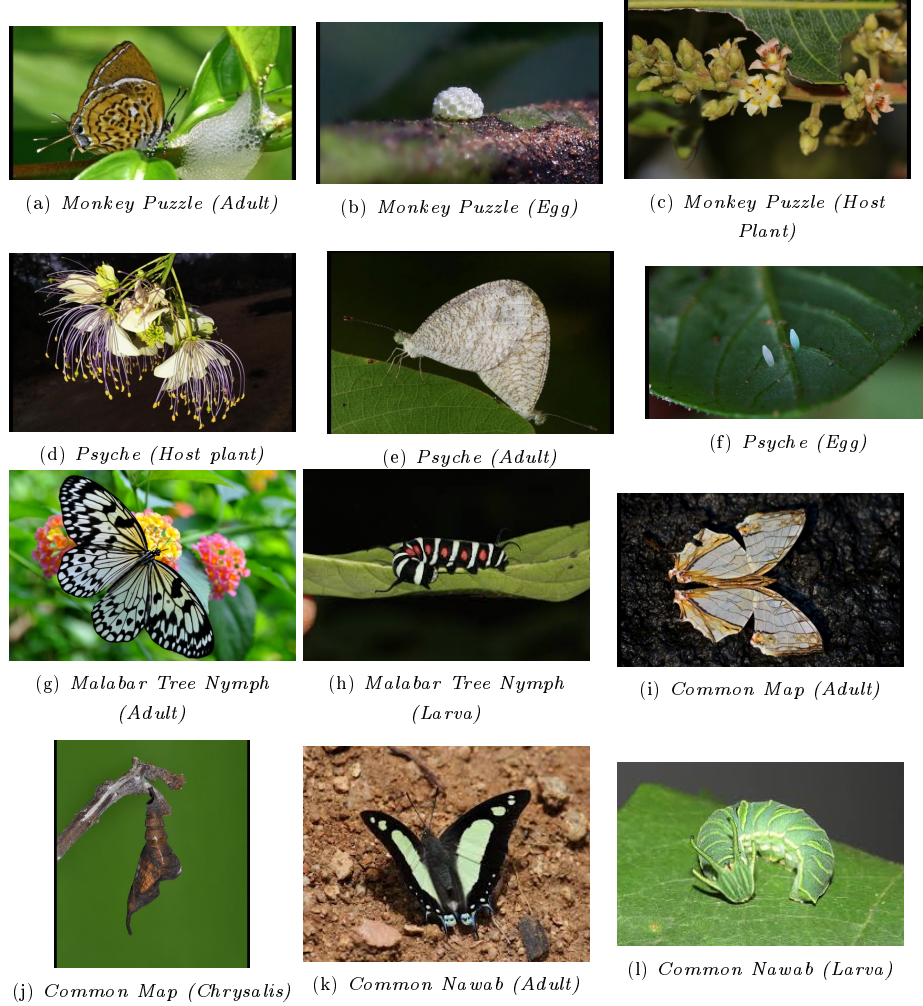


Fig. 2. Multi-stage coverage in IndoLepAtlas across four species. *Rathinda amor* (Monkey Puzzle): adult, egg, host plant; *Leptosia nina* (Common Psyche): adult, egg, host plant; *Idea malabarica* (Malabar Tree Nymph): adult, larva; *Common Map*: adult, chrysalis *Graphium agememnon* (Common Nawab): adult, larva

3.1 Collection and Curation

IndoLepAtlas comprises 61,344 images (60,641 butterflies across 967 species + 703 plant images across 127 larval host plant species), curated primarily from the [Indian Lepidoptera web atlas](#). The full collection and preparation pipeline is formalised in Algorithm 1.

Algorithm 1 IndoLepAtlas Dataset Collection and Preparation Pipeline

Require: Raw species galleries $\mathcal{G}$ from IFoundButterflies.org ▷ Web source

- 1: Download all images and metadata pages per species gallery
- 2: Assign class labels
- 3: Extract per-image metadata via OCR
- 4: Remove non-butterfly entries; resolve taxonomic synonyms
- 5: Separate early life-stage images from adult specimens
- 6: Map GPS coordinates to biogeographic zones
- 7: Encode observation month cyclically
- 8: Assign learned "unknown" embedding where GPS or date is missing
- 9: Perform stratified train/val/test split
- 10: Ensure every species appears in all three splits
- 11: **return** $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{test}}, \mathcal{D}_{\text{early}}$

3.2 Statistics

Table 1. Family distribution (butterflies)

Family	Images	%
Nymphalidae	18,054	29.8
Lycaenidae	16,985	28.0
Hesperiidae	12,361	20.4
Pieridae	6,132	10.1
Papilionidae	5,826	9.6
Riodinidae	1,132	1.9

Table 2. Data splits

Split	Images	%
Train	48,630	79.3
Validation	5,649	9.2
Test	7,065	11.5

3.3 Metadata Description

Location → Biogeographic zone: GPS coordinates mapped to India’s 9 established biogeographic zones (Wildlife Institute of India classification): Western Ghats, Deccan Peninsula, Western Himalayas, Eastern Himalayas, Northeast India, Indo-Gangetic Plain, Semi-Arid, Desert, and Andaman & Nicobar Islands. A 10th "unknown" index is used when GPS is missing. Zone-level encoding is ecologically motivated and robust to typical $\pm$ km GPS errors. Each zone is represented as a learned 32-dimensional embedding vector. Images lacking location or date (cyclically encoded) use a learned "unknown" embedding token during geotemporal fusion.

3.4 Dataset Quality Validation

For automated bbox verification, 100 random images per subset are spot-checked and failures are then re-annotated via CVAT. To maintain taxonomic consistency, species labels are inherited from expert-curated [IFoundButterflies](#) directory structure, cross-referenced against Indian butterfly field guides [16]. Geographic/seasonal distributions are cross-referenced against known species ranges, the confusion matrix analysis confirms systematic confusion occurs between con-generic species.

3.5 Coverage Gap Analysis

Table 3. Comparison with Existing Datasets

Dataset	Spp.	Images	Domain	Loc.	Temp.	Plants	India
CUB-200-2011	200	11,788	Birds	×	×	×	×
iNaturalist	5,089	675,170	Multi	✓	✓	×	×
IP102 [15]	102	75,222	Insects	×	×	×	×
IndoLepAtlas	967	60,641	Butterflies	✓	✓	✓	✓

Table 4. iNat Coverage Gap Analysis

iNat India Records	# Species	% Dataset
0 (absent)	145	15.1%
1–10	249	25.9%
11–50	194	20.2%
51–100	78	8.1%
101–500	147	15.3%
>500	148	15.4%

Cross-referencing IndoLepAtlas species against iNaturalist India research-grade observations reveals that 69.3% of our species have fewer than 100 India-specific iNat records, and 145 species (15.1%) have zero records, confirming the under-representation of Indian Lepidoptera species within global repositories.

4 Experimental Setup

Experiments execute on a DGX V100 cluster in two phases.

- **Backbone:** ConvNeXt-Tiny (ImageNet pretrained)
- **Optimiser:** AdamW, differential learning rates (backbone $0.1\times$, new heads 10^{-4})
- **Scheduler:** Cosine annealing with 5-epoch warmup
- **Training:** Mixed precision (AMP), gradient norm clipping at 5.0
- **Checkpoint selection:** Best macro-F1 on validation (ensures long-tail sensitivity)

- **Augmentation (train):** Random resized crop, flipping, rotation, color jitter, ImageNet normalisation.
- **Augmentation (val/test):** Deterministic resize + centre crop, ImageNet normalisation.
 - Unit I (Loss dynamics): 30 epochs, 4 configs
 - Units II–III (Architecture, Freezing): 40 epochs, 7 configs

5 Baseline Experiments and Results

5.1 Unit I: Long-Tail Loss Dynamics (30 epochs)

Question: How do different loss formulations and sampling strategies impact representational capacity on long-tailed fine-grained distributions?

Table 5. Validation accuracy for loss-sampling configurations.

Configuration	Top-1	Macro F1	Dense Acc	Sparse Acc
CE + Balanced	87.06	85.48	87.67	85.84
CE + Unbalanced	87.60	80.23	91.22	80.35
Focal + Balanced	82.69	82.64	80.85	86.40
Focal + Unbalanced	84.83	82.70	85.70	83.07

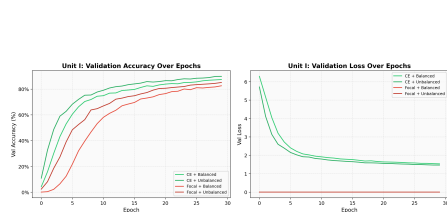


Fig. 3. Unit I training curves.

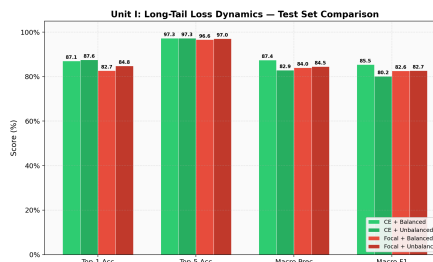


Fig. 4. Unit I metric comparison.

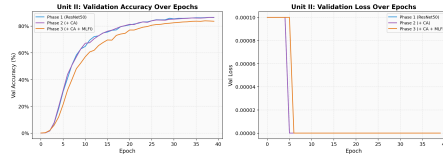
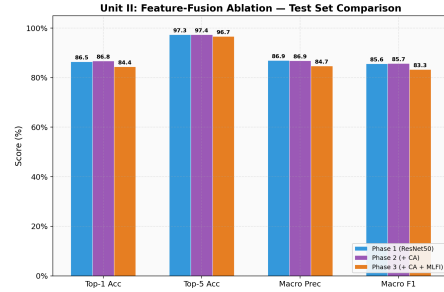
The model abandons rare species to maximise majority-class performance. Instead, inverse-frequency sampling ensures rare-class gradients remain proportional regardless of distribution skew. Explicit sampling works better for the long-tail problem than the loss function.

5.2 Unit II: Feature-Fusion Ablation (40 epochs)

Question: Where is the optimal structural transition point for integrating attention mechanisms, and does multi-level feature integration improve fine-grained classification?

Table 6. Validation accuracy for architectural configurations

Configuration	Top-1	Macro F1	Dense Acc	Sparse Acc
Phase 1 (Baseline)	86.50	85.65	86.36	86.76
+ CA	86.77	85.75	86.13	88.05
+ CA + MLFI	84.43	83.34	83.32	86.66

**Fig. 5.** Unit II training curves.**Fig. 6.** Unit II metric comparison.

Coordinate Attention (CA) preserves spatial morphology. It decomposes channel-spatial attention into two horizontal and vertical pooling — preserving directional structure naturally suited to butterfly morphology (horizontal banding, vertical vein streaking). Unlike SE-Net’s global average pooling, CA retains *where* a pattern occurs. At 40 epochs, MLFI reaches 84.43% Top-1 and 86.66% Sparse—demonstrating that multi-level fusion does benefit rare classes. The newly initialised MLFI layers (Xavier init, full learning rate) create an optimisation imbalance with the near-converged backbone ($0.1 \times$ learning rate), resulting in a poorly conditioned loss landscape.

5.3 Unit III: Transfer Learning Layer Freezing (40 Epochs)

Question: Which hierarchical layers contain domain-agnostic vs domain-specific representations, and what is the optimal adaptation pathway?

Table 7. Layer freezing accuracy

Configuration	Top-1	Macro F1	Dense Acc	Sparse Acc
End-to-End	84.79	84.11	83.89	86.61
Head Only	53.20	57.02	44.46	70.75
Freeze Early	83.77	83.20	82.74	85.84
Freeze Late	84.33	84.02	83.15	86.71

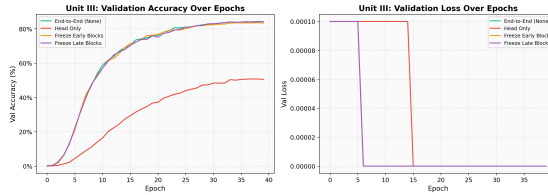


Fig. 7. Unit III training curves.

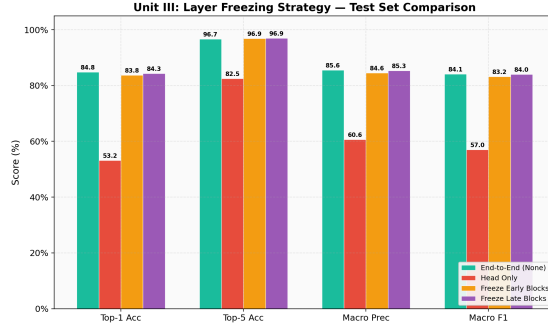


Fig. 8. Unit III metric comparison.

ImageNet activations encode “furry surface textures” and “rounded shapes,” not “submarginal band” or “discal spot.” Fine-grained biological classification is fundamentally a *domain-shift problem disguised as a classification problem*. Freezing late blocks (84.33%) slightly outperforms freezing early blocks (83.77%), implying early-layer Gabor-like edge detectors require significant recalibration for butterfly wing scales and biological venation. We also observe that full gradient backpropagation achieves the best macro-F1 (84.11%) and most balanced stratum performance (Dense 83.89%, Sparse 86.61%).

6 Model Comparison

To contextualise our ablation findings and evaluate our geotemporal fusion module, we benchmark six architectures on the same dataset split (Table 8).

6.1 Main Comparison

Table 8. Model comparison on the test set (40 epochs, CE loss, balanced sampling).

Model	Params	Top-1	Top-5	Macro P	Macro F1	Wt. F1
ResNet-101	44.5M	48.76	76.91	49.94	47.60	47.76
EfficientNet-B5	30.3M	86.96	97.05	87.33	85.92	86.57
ViT-B/16	86.5M	89.99	97.51	90.14	88.11	89.51
MLFI Warmup	40.8M	87.20	97.34	87.84	85.72	86.76
Geo-Shuffled*	40.8M	86.36	97.12	86.41	84.57	85.86
Geotemporal (ours)	40.8M	89.38	98.01	89.13	87.37	88.94

*Geo-Shuffled: zone/month labels randomly permuted at inference to ablate geotemporal signal.

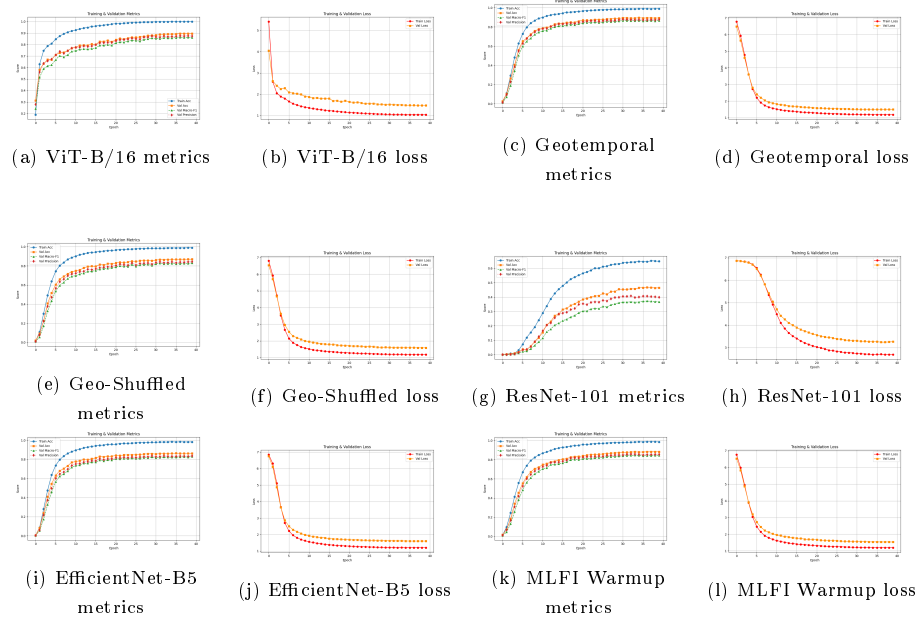


Fig. 9. Training metrics and loss curves for all six model configurations. (a–d) ViT-B/16 and Geotemporal (ours). (e–h) Geo-Shuffled and ResNet-101. (i–l) EfficientNet-B5 and MLFI Warmup.

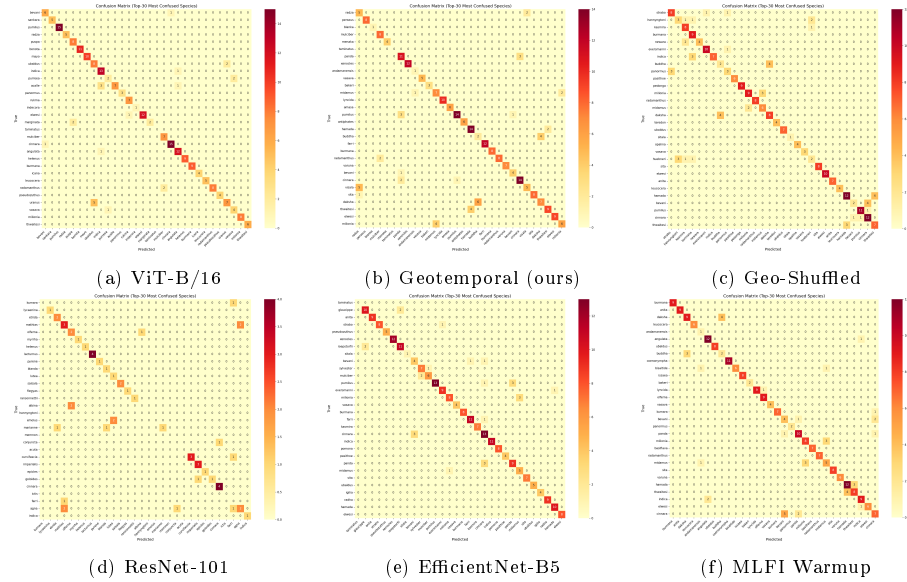


Fig. 10. Confusion matrices for all six model configurations. Confusion is concentrated among congeneric species pairs, supporting the geotemporal disambiguation hypothesis.

6.2 Per-Stratum Analysis

Table 9. Per-stratum analysis (% accuracy and F1).

Model	Dense Acc	Dense F1	Sparse Acc	Sparse F1	Δ Acc
ResNet-101	45.02	22.53	56.29	45.80	+11.27
EfficientNet-B5	87.33	68.30	86.20	75.07	-1.13
ViT-B/16	91.60	79.40	86.76	75.18	-4.84
MLFI Warmup	88.36	73.07	84.86	72.84	-3.50
Geo-Shuffled	87.87	72.32	83.32	71.89	-4.55
Geotemporal	90.35	76.74	87.43	75.38	-2.92

6.3 Confusion Matrix Analysis

Table 10. Top confusion pairs per model.

Model	True Species	Predicted As	N
ResNet-101	<i>Symbrenthia brabira</i>	<i>S. lilaea</i>	7
ResNet-101	<i>Eurema blanda</i>	<i>E. hecabe</i>	7
EfficientNet-B5	<i>Azanus ubaldus</i>	<i>A. uranus</i>	6
EfficientNet-B5	<i>Tarucus indica</i>	<i>T. nara</i>	5
ViT-B/16	<i>Tapena thwaitesi</i>	<i>Taraka hamada</i>	8
ViT-B/16	<i>Azanus uranus</i>	<i>A. ubaldus</i>	5
Geotemporal	<i>Papilio daksha</i>	<i>P. buddha</i>	6
Geotemporal	<i>Mycalesis visala</i>	<i>M. radza</i>	5

6.4 Key Findings

1. **Geotemporal fusion** is competitive with **ViT-B/16**. Achieves 89.38% Top-1 with $2.1\times$ fewer parameters. It achieves the **best Top-5 accuracy (98.01%)** across all models.
2. **Geotemporal metadata provides a genuine signal**. The geo-shuffled control (same architecture, randomised location/month) drops by: -3.02% Top-1 ($89.38 \rightarrow 86.36$) and -2.80% Macro F1 ($87.37 \rightarrow 84.57$). The geotemporal branch learns meaningful biogeographic priors, does not just add parameters.
3. **Best sparse-class performance from geotemporal model**. 87.43% accuracy on sparse classes (<50 training images), outperforming all baselines including ViT-B/16 (86.76%). The Dense-Sparse gap narrows to $\Delta = -2.92$ vs ViT-B/16’s $\Delta = -4.84$, suggesting geotemporal priors help disambiguate underrepresented species.
4. **MLFI warmup provides modest gains**. 87.20% vs 86.36% (geo-shuffled), but underperforms full geotemporal fusion (89.38%), suggesting multi-level feature integration benefits from geospatial context.

7 Discussion

Our study highlights why architectural design and ecological and temporal context are critical for fine-grained biological classification. Three structural constraints drive the primary failure modes:

- **Spatial Collapse** Coordinate Attention addresses spatial collapse by preserving the directional structure of butterfly morphology (e.g., horizontal bands vs. vertical veins). Adding CA requires <0.5% more parameters but improves sparse-class accuracy by 1.29 points.
- **Texture Loss** MLFI preserves essential early-layer texture features, but requires careful optimisation. Applying a learning rate warmup to newly initialised branches recovers a **2.77-point Top-1 improvement** over the unwarmed baseline (87.20% vs. 84.43%).
- **Domain Shift** The failure of Head-Only fine-tuning (53.20%) highlights a severe domain shift from natural images to biological specimens. ImageNet features encode object silhouettes rather than critical fine-grained features like submarginal bands, making full end-to-end recalibration essential for transferability.
- **Geotemporal Signal** Beyond architecture, geotemporal fusion bridges the remaining accuracy gap by encoding biological reality. The most visually confused species pairs (*Kallima inachus* ↔ *Kallima horsfieldii*, *Malabar Rose* ↔ *Common Rose*, *Mycalesis francisca* ↔ *Mycalesis gotama*) are congeneric species with overlapping wing patterns but **distinct geographic ranges and flight seasons**—the additional context the geotemporal embeddings provide for disambiguation.

8 Conclusion

IndoLepAtlas fills a documented gap in the insect classification literature as the first India-specific Lepidoptera dataset with integrated geotemporal metadata, early life-stage documentation, and host plant associations. By establishing a 60,641-image baseline covering 967 species, we aim to bridge a major gap in region-based ecological classification.

Our evaluations demonstrate that modern architectures require targeted adaptations for biological classification. Standard ImageNet transfer fails due to severe domain shifts, making end-to-end fine-tuning mandatory. However, combining Coordinate Attention with balanced sampling creates an efficient baseline that mitigates both spatial collapse and long-tail imbalance. Crucially, our geotemporal fusion module proves that integrating biogeographic and seasonal context provides genuine discriminative signal, bridging the accuracy gap to match much larger Vision Transformers at half the parameter cost.

8.1 Future Work

- **Prototypical network heads** for the 533 classes with <50 images, enabling few-shot classification of rare species.

- **Domain-specific self-supervised pretraining** (DINO/MAE on butterfly images) to bootstrap features from unlabelled in-domain data, reducing dependence on ImageNet initialisation.
- **Host plant association features** for habitat-aware species prediction
- **Early stage detection** for conservation and finer species identification.
- **Hierarchical classification** exploiting the taxonomic tree (family → genus → species) to reduce confusion between congeneric species.

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